Shubham Khanapure

Senior AI/ML Engineer | Generative AI, Agentic Systems & Production ML

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SUMMARY

AI/ML engineer with **7 years** taking machine learning from problem framing to production ownership. Builds LLM and agentic systems — specialized agents, orchestrators, tool calling, RAG, guardrails, human-in-the-loop review — on top of computer vision, graph-based reasoning and time-series forecasting, served as containerized Python APIs with evaluation, monitoring and CI/CD. Currently owns an end-to-end P&ID digitization and Functional Specification generation platform (**82% detection accuracy**) and an S&OP demand-forecasting agent. Microsoft Certified Azure AI Engineer, hands-on with Azure ML, Azure AI Foundry and secured enterprise cloud.

CORE SKILLS

Generative AI and Agents: LangGraph, LangChain, LlamaIndex, Azure OpenAI, Azure AI Foundry, GPT-4, multi-agent orchestration, specialized agents, tool calling, prompt engineering, RAG and agentic RAG, structured output, guardrails, human-in-the-loop, LLM evaluation, LoRA / QLoRA fine-tuning

Machine Learning and Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, BERT, scikit-learn, XGBoost, transfer learning, model evaluation and error analysis

Computer Vision and Document AI: YOLO (v8 / v11), Detectron2, SAHI tiled inference, OpenCV, template matching, OCR (DocTR, PaddleOCR, Tesseract), Azure Document Intelligence, Azure AI Language, NER, text classification

Time Series and Forecasting: SARIMA, SARIMAX with exogenous features, TimeGPT, Prophet, statsmodels, S&OP demand and supply planning, backtesting, MAPE / RMSE evaluation

MLOps and Serving: MLflow, Azure ML jobs and pipelines, model registry and versioning, Docker, Kubernetes, Azure DevOps CI/CD, FastAPI inference services, monitoring and drift detection, ONNX, TensorRT, CUDA, mixed-precision training, quantization

Engineering and Data: Python, FastAPI, Django, Flask, REST APIs, microservices, SQL, PostgreSQL, MongoDB, graph databases (Neo4j, graph traversal), vector databases (Pinecone, ChromaDB, Weaviate), Azure AI Search, PySpark, Azure Databricks, Pandas, NumPy, Git, pytest

Cloud and Security: Azure ML workspaces, Web Apps, Functions, Blob Storage, Key Vault, Container Registry, private endpoints, virtual network isolation, managed identity, RBAC, PII detection and redaction

EXPERIENCE

AI/ML Engineer — Generative AI, Agentic Systems & Computer Vision

Emerson Export Engineering Centre, Pune | Apr 2025 – Present

Functional Specification Generation (FSG) — P&ID digitization to auto-authored specifications

- **Object detection:** Built a hybrid symbol-detection stage for Piping and Instrumentation Diagrams combining YOLOv8/v11, Azure Custom Vision and template matching with SAHI tiled inference for small symbols, reaching **82% accuracy** on high-resolution schematics; trained on CUDA GPUs with mixed precision.
- **Text extraction:** Developed a separate OCR stage (DocTR, PaddleOCR, Azure Document Intelligence) that extracts instrument tags, line numbers and annotations from dense drawings.
- **Symbol-to-tag mapping:** Implemented bounding-box geometry and spatial-association algorithms that map each detected symbol to its correct text tag, converting unstructured drawings into structured records.
- **Graph modeling and traversal:** Converted mapped entities into a graph-database representation of the P&ID (instruments, lines, valves, equipment as nodes and connections as edges) and built traversal and validation logic that reconstructs and verifies complete end-to-end process flows.
- **Agentic document generation:** Designed a LangGraph multi-agent system — specialized agents with dedicated prompts and tools under an orchestrator — that consumes the validated graph flow and auto-authors the detailed Functional Specification as a structured **.docx**, section by section, covering each stage of the instrumentation pipeline; replaced multi-day manual authoring with automated drafting plus reviewer sign-off.
- **Quality and safety:** Added structured-output validation, guardrails, an automated evaluation harness for factuality and standards compliance, and an AI-Shield service for PII detection and instrumentation-domain sanitization with a human-in-the-loop review queue.

S&OP Demand Forecasting & Platform Engineering

- **Forecasting models:** Built an **18-month** rolling Sales and Operations Planning (S&OP) forecast of incoming client demand using SARIMA and SARIMAX with exogenous supply and order features, benchmarked against TimeGPT, translating demand into required resource counts against present supply data.
- **S&OP agent:** Delivered an agentic chatbot that answers planning questions in natural language over the forecast outputs — demand trend, resource requirement and supply-gap queries — with tool-calling access to the forecasting models and planning data.
- **Platform and MLOps:** Shipped the FastAPI backend and React frontend with role-based access control and Entra ID SSO; owned Azure ML training jobs, MLflow tracking and registry, Dockerized inference services, Azure DevOps CI/CD, and secured infrastructure with private endpoints, VNet isolation and Key Vault.

Data Scientist — Automation & Data Engineering
TEKsystems India (Client: HSBC), Pune | Jul 2024 – Apr 2025

- **Pulse Intelligence — end-to-end module automation:** Automated an entire production tracking module, including portal automation and automated capture and analysis of **app trace, sys trace and flow trace** data, removing manual trace collection and triage.
- **Data pipelines:** Built end-to-end Azure Databricks pipelines (PySpark) for ingestion, transformation and analysis of large-scale trace and document data with GPU-accelerated inference stages.
- **Production application:** Developed and ran the backend and desktop application in production, with structured logging, alerting, pytest regression suites and Azure DevOps CI/CD on Docker for weekly releases.
- **Document understanding:** Engineered an OCR pipeline (EAST/CRAFT detection with Tesseract, PaddleOCR and DocTR recognition) for large-scale financial document processing.

Associate Consultant — Data Scientist
KPMG Global Services, Pune | Jun 2022 – Jul 2024

- Built enterprise generative AI products on GPT-4, Azure OpenAI and LangChain — RFP Response Generator, CSRD/SEC Readiness Tool, Job Architecture Optimizer — cutting manual effort **70–80%** for business teams.
- Designed the semantic-search and RAG architecture (sentence transformers, Pinecone/Weaviate) behind those products, served from Django and FastAPI services on auto-scaling Azure Web Apps with A/B rollout.
- Applied LoRA/QLoRA fine-tuning for cost and latency gains; established MLflow retraining, model versioning and production monitoring. Awarded **KPMG Megastar (2024)** and **Encore Kudos (2023)**.

Engineer — Data Scientist
Vyom Labs, Pune | Sep 2019 – May 2022

- Built an automated ticket-classification and routing system with BERT and spaCy pipelines (NER, text classification) reaching **88% accuracy** in production.
- Developed supervised and unsupervised models (PyTorch, scikit-learn, XGBoost) and Apache Airflow pipelines with dimensionality reduction for feature engineering.

EDUCATION & CERTIFICATIONS

B.E. Computer Science — Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Pune | 2015 – 2019

Microsoft Certified: Azure AI Engineer Associate (AI-102) | Deep Learning Specialization, Coursera | Data Scientist Nanodegree, Udacity